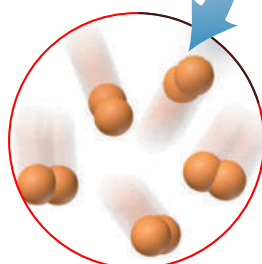


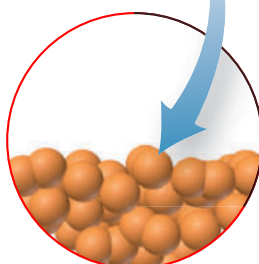


UNIT 05

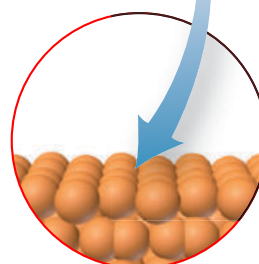
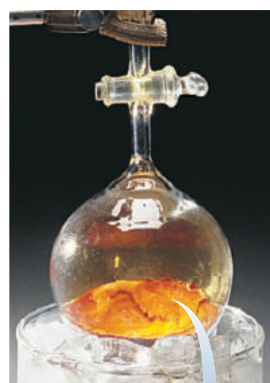
PHYSICAL STATES OF MATTER



Gas: Particles are far apart, move freely, and fill the available space.



Liquid: Particles are close together but move around one another.



Solid: Particles are close together in a regular array and do not move around one another.

GAS: Gas is that state of matter which has no definite shape and they has no definite volume.

A gas consists of molecules separated wide apart in empty space. The molecules are free to move about throughout the container.

GENERAL CHARACTERISTICS OF GASES:

1. EXPANSIBILITY: Expansibility is the ability of a gas to expand and occupy the entire volume of its container when the temperature or pressure changes.

► Gases have limitless expansibility. They expand to fill the entire vessel they are placed in.

2. COMPRESSIBILITY: Compressibility is the ability of a gas to decrease in volume under pressure.

► Gases are easily compressed by application of pressure to a movable piston fitted in the container.

► Liquids and solids are not like that. They are essentially incompressible.

3. DIFFUSIBILITY: Diffusibility is the ability of a gas to spread and mix evenly with other gases.

► Gases can diffuse rapidly through each other to form a homogeneous mixture.

4. PRESSURE: Pressure is the force exerted by gas particles when they collide with the walls of their container.

► Gases exert pressure on the walls of the container in all directions.

When we blow air into a round balloon, the balloon becomes spherical and uniform. The air in the balloon obviously pushes out equally in all directions. If we were to fill the balloon with water instead (for scientific reasons only, of course), we would notice that the balloon sags. The water accumulates in the bottom of the balloon.

THE KINETIC MOLECULAR THEORY:

K.E is the summarized set of assumptions collectively known as the kinetic molecular theory, or simply the kinetic theory. The major points of this theory as applied to gases are as follows:

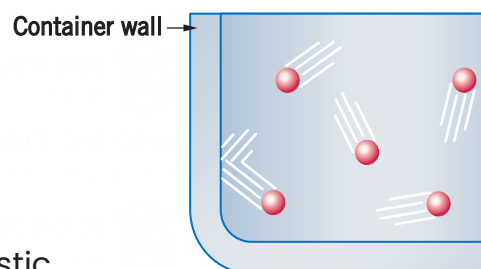
1. SPACES: A gas is composed of very small particles called molecules, which are widely spaced and occupy negligible volume. A container of gas is thus mostly empty space.

2. MOTION: The molecules of a gas are in rapid, random motion, colliding with each other and the sides of the container. Pressure is a result of these collisions with the sides of the container.

3. COLLISION: All collisions involving gas molecules are elastic.

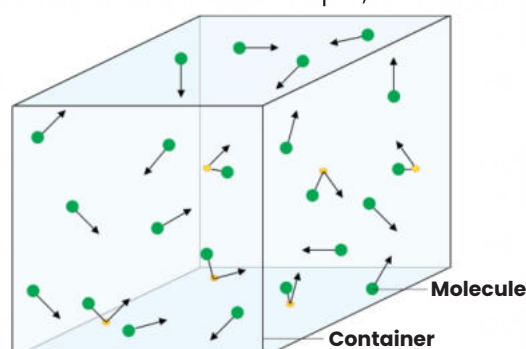
4. ATTRACTION OR REPULSION: Gas molecules have negligible attractive (or repulsive) forces between them.

5. TEMPERATURE & K.E: The temperature of a gas is related to the average kinetic energy of the gas molecules. Also, at the same temperature, different gases have the same average kinetic energy (K.E.).



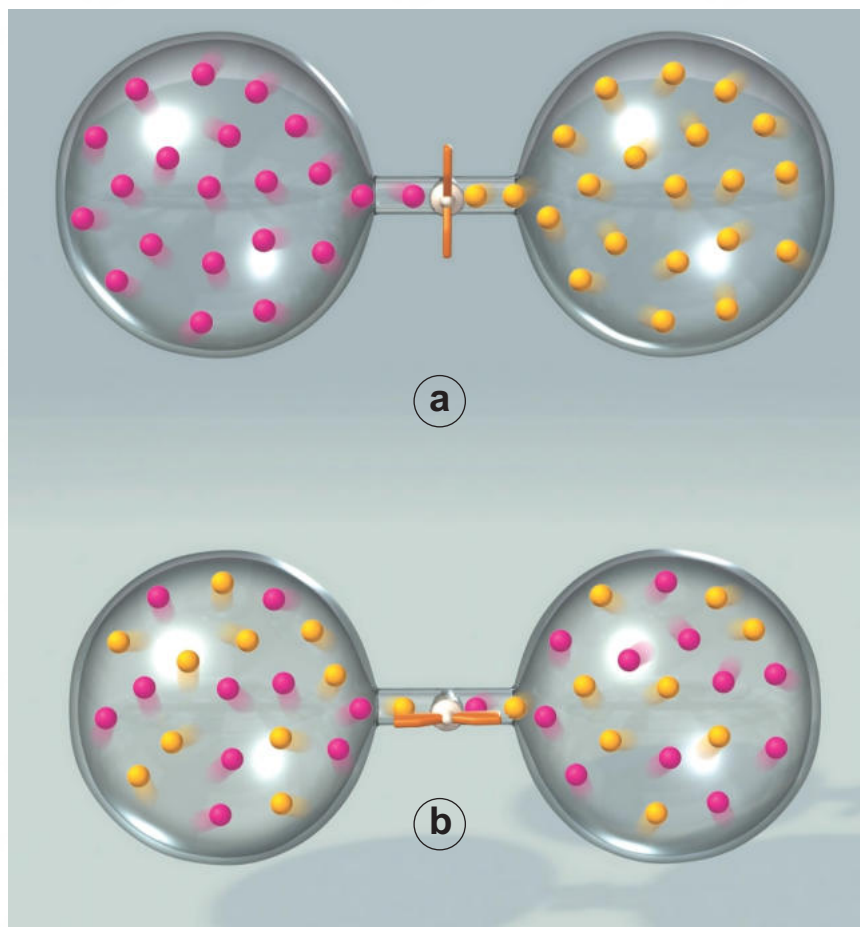
Gas molecule =

▲ **Gases** Gas molecules are in rapid, random motion.



DIFFUSION:

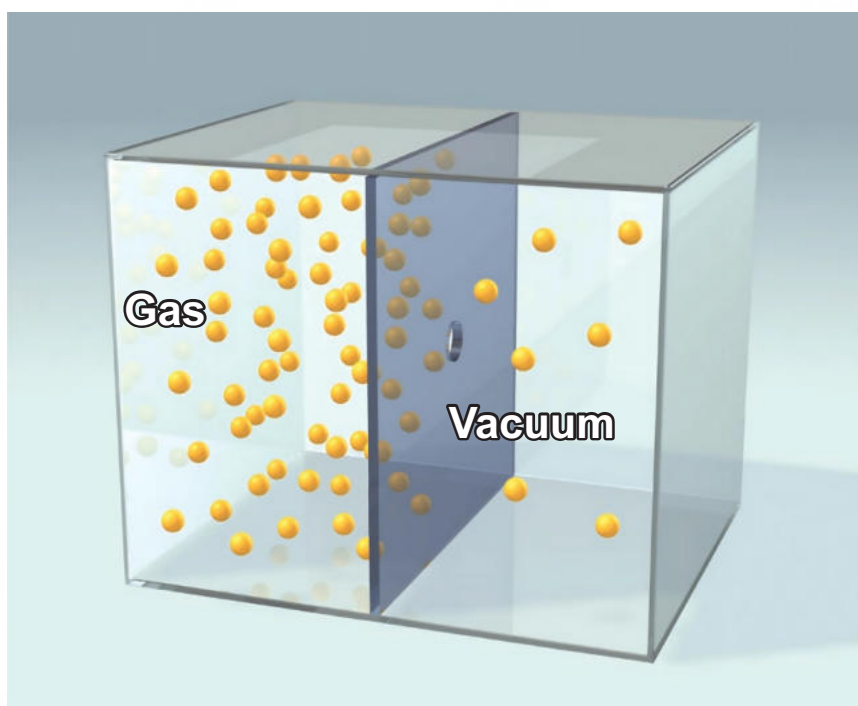
"Spreading of a substance is diffusion or the movement of a substance from their higher concentration to lower concentration."



◀ Diffusion is the mixing of gases. **(a)** Two different gases in separate containers. **(b)** When the stopcock is opened, the gases mix by diffusion.

EFFUSION:

"When a gas escapes through a pin-hole into a region of low pressure or vacuum, the process is called Effusion."



▲ Effusion is the escape of a gas into a vacuum.

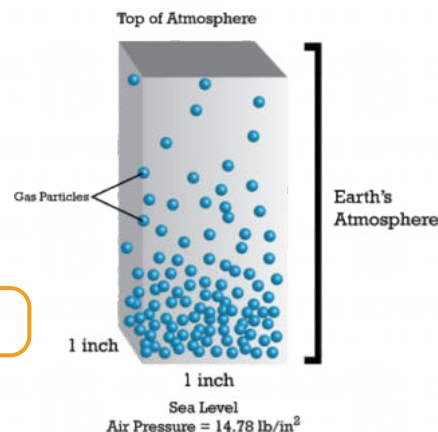
PRESSURE:

"Force per unit area is pressure."

$$P = \frac{F}{A}$$

UNITS: The SI unit of pressure is N m^{-2} . It is also called Pascal (Pa)

$$1 \text{ atm} = 760 \text{ torr} = 760 \text{ mm of Hg} = 101325 \text{ pascal or } \text{Nm}^{-2}$$



MOBILITY:

The ability of a substance to move freely.

DENSITY:

Mass per unit volume is density or actually it tells the degree of compactness of any matter.

GAS LAWS:

PRESSURE-VOLUME LAW (BOYLE'S LAW):

INTRODUCTION:

The relationship between the volume and pressure of a sample of gas under constant-temperature conditions was first described in 1660 by the English scientist Robert Boyle.

STATEMENT:

"The volume of a given mass of a gas is inversely proportional to its applied pressure if the temperature is held constant."

EXPLANATION AND DERIVATION:

The inverse relationship suggested by Boyle is represented as;

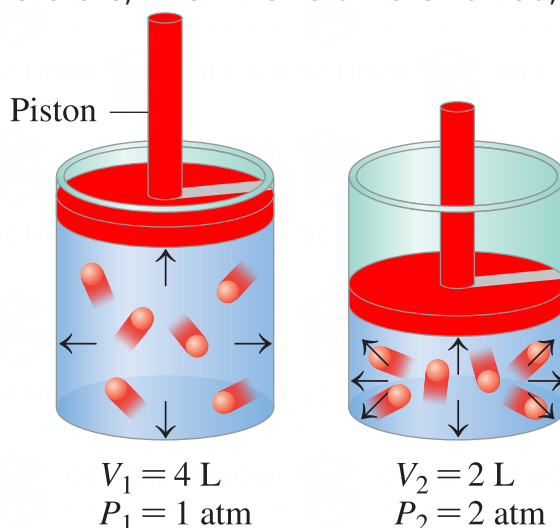
$$V \propto \frac{1}{P}$$

or as an equality with k the constant of proportionality:

$$V = k/P \text{ or } PV = k$$

The image given below illustrated Boyle's Law, which states that for a fixed amount of gas at constant temperature, the pressure and volume are inversely proportional. As the volume decreases (from 4 L to 2 L), the pressure increases (from 1 atm to 2 atm). The gas particles collide more frequently with the container walls as the volume is reduced, increasing the pressure. Therefore, when the volume is halved, the pressure doubles.

$$P_1V_1 = P_2V_2 = k$$



LINK TO LEARNING

Watch this video for the visual understanding of boyles's law (<https://youtu.be/Vj2x1tAgdjs>)

LINK 01



TEMPERATURE-VOLUME LAW (CHARLES LAW):

INTRODUCTION: In 1787, Jacques Charles, a balloonist as well as a physicist, proposed that the volume of a gas is related to the temperature. This proposal became Charles's law.

STATEMENT: "The volume (V) of a gas is directly related to the absolute temperature (T) at constant pressure (P) and amount (n) of gas."

LINK TO LEARNING

This video (<https://youtu.be/ZgTTUuJZAFs>) shows how cooling and heating a gas causes its volume to decrease or increase, respectively. Another video (<https://www.youtube.com/watch?v=Y4TEgYSqKCM>)

LINK 01



LINK 02

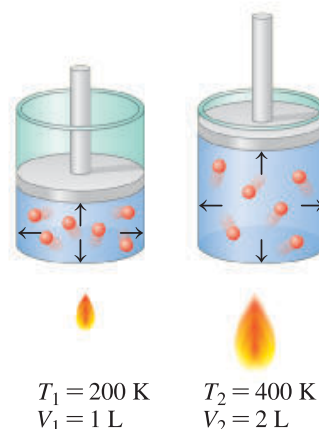


DERIVATION:

$$V \propto T \quad \text{or} \quad \frac{V}{T} = k$$

This image demonstrates Charles's Law, which states that at constant pressure, the volume of a gas is directly proportional to its temperature. As the temperature increases (from 300 K to 400 K), the gas expands, increasing the volume (from 1 L to 2 L). The gas particles move faster at higher temperatures, pushing the walls of the container outward, which causes the volume to increase. Therefore, when the temperature doubles, the volume also doubles.

For a quantity of gas at two temperatures at the same pressure,



The Relationship between Temperature and Volume



(a) Air-filled balloons are supercooled with liquid nitrogen.



(b) The balloons collapse.



(c) The balloons expand when the air inside warms up.

critical thinking

Explain Use the kinetic-molecular theory to explain what happens to the balloons.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

LIQUID STATE:

"The state of matter which have no definite shape but they have the definite volume." A liquid has molecules touching each other. However, the intermolecular space, permit the movement of molecules throughout the liquid.

TYPICAL PROPERTIES OF LIQUID:

Evaporation:

"The process of changing of a liquid into a gas phase or vapours is called evaporation."

Factors affecting:

(i) Temperature: At high temperature, rate of evaporation is high because at high temperature kinetic energy of the molecules increases so high that they overcome the intermolecular forces and evaporate rapidly.

(ii) Surface area: Evaporation is a surface phenomenon so, greater is surface area, greater is evaporation and vice versa.

(iii) Intermolecular forces: If intermolecular forces are stronger, molecules face difficulty in evaporation.

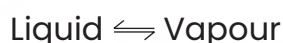
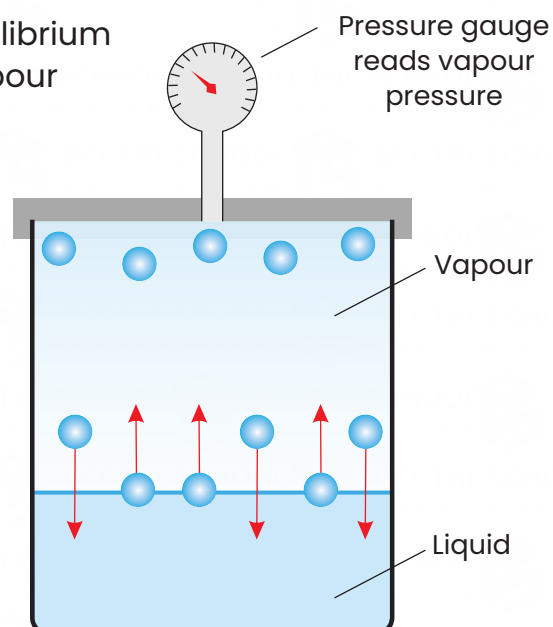
Vapour pressure:

"The pressure exerted by the vapours of a liquid at equilibrium with the liquid at a particular temperature is called vapour pressure of a liquid."

Explanation:

If the liquid is placed in a closed vessel, the molecules with high kinetic energies escape into space above the liquid. As the number of molecules in the gas phase increases, some of them strike the liquid surface and are recaptured (condensation).

A stage comes when the number of molecules escaping from the liquid is equal to the number of molecules returning to the liquid. In other words, the rate of evaporation exactly equals the rate of condensation. Thus, a dynamic equilibrium is established between the liquid and the vapour at the given temperature.



FACTORS AFFECTING:

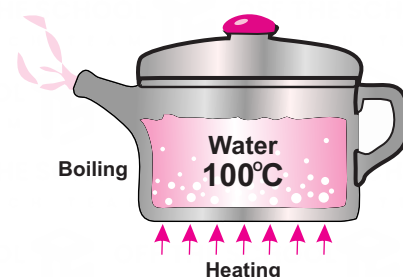
(i) Nature of liquid: Vapour pressure depends upon the nature of liquid. Polar liquids have low vapour pressure than non-polar liquids at the same temperature.

(ii) Size of molecules: Small sized molecules can easily evaporate than big sized molecules hence, small sized molecular liquids exert more vapour pressure

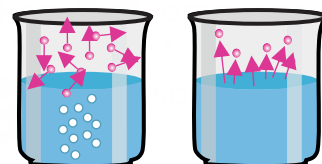
(iii) Temperature: At high temperature, vapour pressure is higher than at low temperature. Vaporize and exert pressure.

BOILING POINT:

"The normal boiling point of a liquid is the temperature at which the vapor pressure is equal to 760 torr or 1 atm." Boiling occurs when a liquid is heated to its boiling point, causing vaporization to happen throughout the liquid, forming bubbles. Evaporation,



however, happens at the surface of the liquid and can occur at any temperature below the boiling point, where only the surface molecules escape into the air.



Boiling

Evaporation

FREEZING POINT:

“The temperature at which the vapour pressure of a liquid state becomes equal to the vapour pressure of its solid state is known as Freezing Point of a liquid.”

FACTORS AFFECTING:

- i. Temperature.
- ii. Imfs.

CONDENSATION:

The change from the gas phase to the liquid is called condensation.



▲ Condensed water droplets on glass of windows.

SOLID STATE OR CONDENSED STATE OF MATTER:

The state of matter which have definite shape and volume.

PROPERTIES OF THE CONDENSED OR SOLID STATE:

1. **Density:** They have high density.
2. **Compressibility:** They are essentially incompressible.
3. **Expansibility:** All solids expand on heating.
4. **Shape and volume:** They have a fixed shape and volume.

THE CONDENSED STATES AND KINETIC THEORY:

1. **Forces of attraction:** The molecules in solids are closely packed due to stronger forces of attraction.
2. **Intermolecular space:** The molecules are unable to move freely as they have little space between them.
3. **Motion:** The molecules can vibrate and rotate in their fixed position.
4. **Shape and volume:** Solids have definite shape and definite volume.
5. **Melting point:** Pure solids have sharp melting point.

Rigidity: The solids are closely packed due to this solid are not mobile. Therefore, solids have rigidity in their structure.

Density: solids are much denser than liquid or gas.

SUBLIMATION: Some solids can transition directly into the gaseous state, bypassing the liquid state, via a process known as sublimation.

At room temperature and standard pressure, a piece of dry ice (solid CO_2) sublimates, appearing to gradually disappear without ever forming any liquid.



▲ Under atmospheric conditions, solid carbon dioxide does not melt; it can only **sublime**.

DEPOSITION: The reverse of sublimation is called deposition, a process in which gaseous substances condense directly into the solid state, bypassing the liquid state. The formation of frost is an example of deposition.

► **Sublimation of Iodine.**



TYPES OF SOLID:

Amorphous and Crystalline Solids:

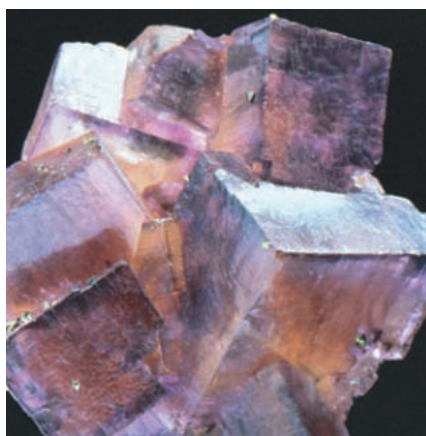
There are basically two general categories of solids: amorphous and crystalline.

(i) Amorphous solids are so named because they have no defined shape. The basic particles in amorphous solids are not located in any particular positions. Examples of amorphous solids are glass, rubber, and many plastics.

(ii) In crystalline solids, the molecules or ions are arranged in a regular, symmetrical structure called a crystal lattice. Examples of crystalline solids are salt crystal, a piece of quartz, and many minerals found on Earth naturally form solids with definite or characteristic symmetrical geometric.



Aquamarine



Fluorite



Dolomite

▲ **Crystalline Solids** The minerals shown here are crystalline solids that are ionic compounds. Their symmetrical shapes reflect the ordered geometry of the ions within.

ISOTROPY:

The property of the crystal in which physical properties are same in all directions called isotropy.

ANISOTROPY:

The property of the crystal in which physical properties are different in different directions called anisotropy. For example: graphite conduct electricity parallel to plane not perpendicularly.

CRYSTALLINE SOLIDS	AMORPHOUS SOLIDS
SHAPE	
1. They have definite geometrical shape.	They have indefinite geometrical shape.
ARRANGEMENT	
2. They have highly ordered three dimensional arrangements of particles.	In these solids particles are randomly arranged in three dimensions.
MELTING AND BOILING POINTS	
3. They have sharp m.p and b.p.	They don't have sharp m.p and b.p.

SYMMETRY

4. Crystalline solids are symmetrical.

Amorphous solids are unsymmetrical.

CLEAVAGE

5. Crystalline solids break along particular direction at fixed cleavage planes.

Amorphous solids do not break at fixed cleavage planes.

ALLOTROPY:

"The phenomenon of existence of the element in more than one crystalline form is Allotropy and the forms are allotropes or allotropic forms or modification."

ALLOTROPIC FORMS OF CARBON:

i. Diamond: Diamond is a very hard substance that forms deep inside the Earth under high pressure and temperature from graphite, another form of carbon. Even though they look different, diamond and graphite are made of the same element, carbon.

ii. Graphite: Graphite is made of carbon atoms arranged in flat layers. These layers can slide over each other, which is why graphite is used as a lubricant and in pencil "lead."

iii. Bucky balls: Buckyballs, or Buckminsterfullerene, are a newer form of carbon discovered in 1985. The carbon atoms are arranged in a spherical shape, like a soccer ball. Buckyballs come in different sizes, with the most common being C₆₀ and C₇₀.

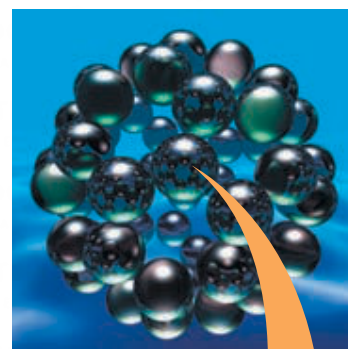
Previous Year's Questions

What is Allotropy? Describe different allotropic forms of carbon.

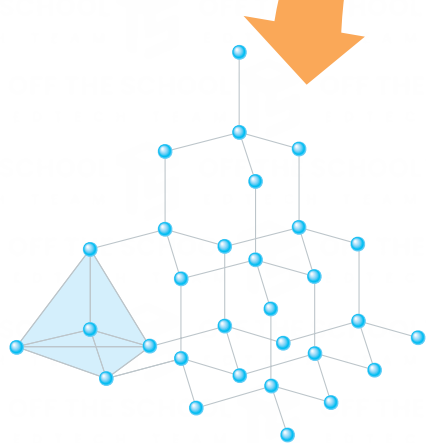
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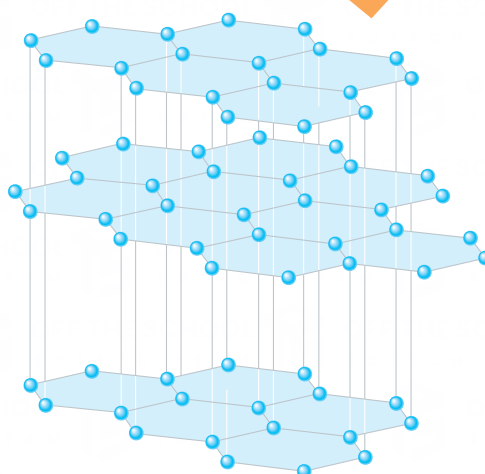
Diamond and graphite are allotropes of carbon.



Bucky Balls.



Diamond



Graphite



Buckminsterfullerene

▲ The Allotropes of Carbon There are three allotropes of carbon diamond, graphite and buckminsterfullerene.

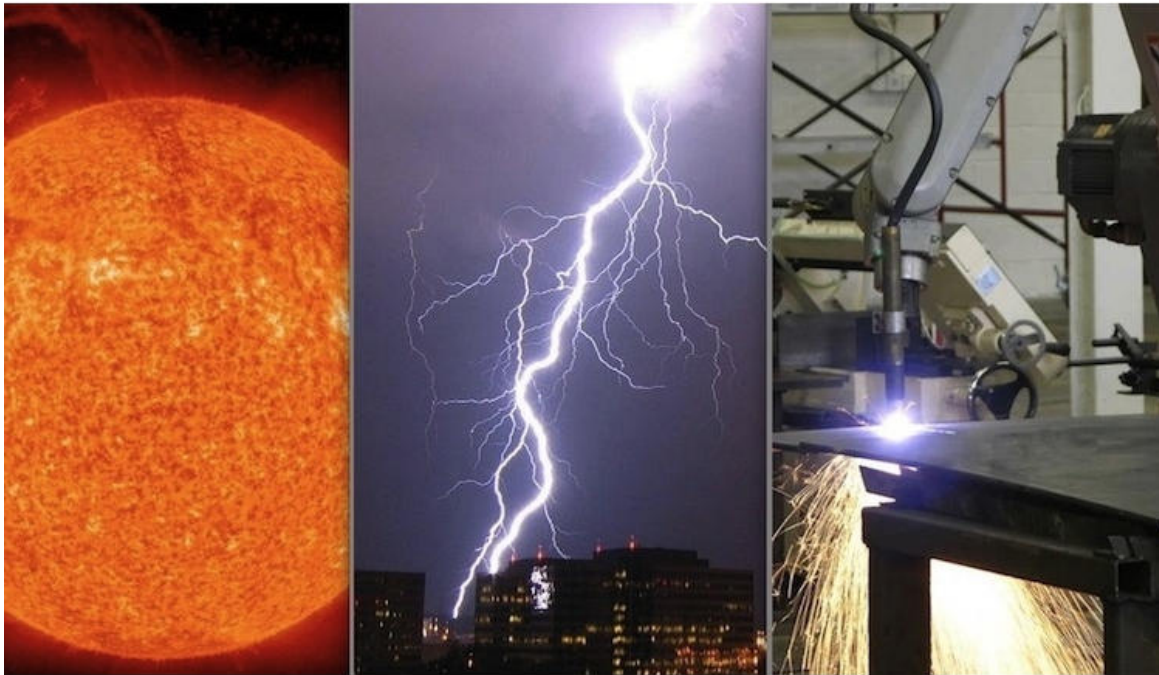
MELTING AND MELTING POINT:

"The co-existence of solid and liquid is melting or some time it's called fusion and the temperature at which solid starts to melt is melting point."

PLASMA: (non-evaluative)

This is the 4th state of matter most of the universe consists of the matter in plasma state.

- A plasma is an extremely ionized gas.
- A plasma is a very good conductor of electricity and is affected by magnetic fields.
- Plasma, like gases have an indefinite shape and an indefinite volume.



BOSE-EINSTEIN CONDENSATES:

Two scientists Satyendra Bose and Albert Einstein discovered another state of matter in 1920 but they did not have the equipment and facilities to make it happen at that time. Afterwards it was discovered in 1995, Bose-Einstein condensates were made with the help of the advancements in technology.

- Carl Weiman and Eric Cornell used magnets and lasers to cool rubidium gas to nearly absolute zero.
- At this super cold temperature, the atoms stop moving much and start sticking together to form a "super-atom."
- Light moves slower through a BEC, which helps scientists learn more about how light works.
- BECs also behave like a superfluid, which means they can flow without any resistance.



NUMERICAL PROBLEMS

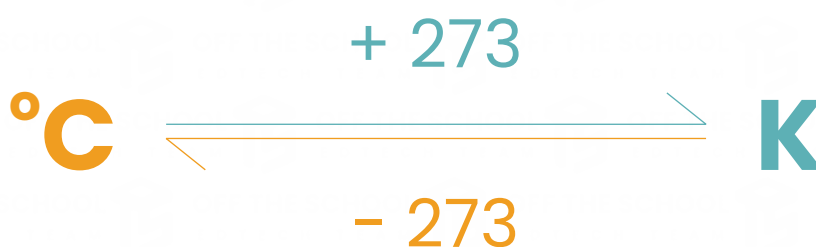
Boyle's Law Numericals ($P_1V_1 = P_2V_2$)

1. A gas occupies a volume of 6 L at a pressure of 2 atm. If the pressure is increased to 4 atm, what will be the new volume, assuming the temperature remains constant?
2. A gas occupies a volume of 10 L at a pressure of 1.5 atm. If the volume is reduced to 5 L, what will be the new pressure?
3. A gas occupies a volume of 4 L at a pressure of 3 atm. What will the volume be when the pressure is reduced to 1 atm?
4. A gas at a pressure of 5 atm has a volume of 8 L. What will be its volume if the pressure is changed to 2 atm?
5. A gas at a pressure of 10 atm occupies a volume of 3 L. What will its volume be when the pressure is reduced to 5 atm?

Charles's Law Numericals ($V_1/T_1 = V_2/T_2$)

1. A gas occupies 4 L at a temperature of 300 K. If the temperature is increased to 400 K, what will be the new volume, assuming the pressure remains constant?
2. A gas occupies a volume of 6 L at a temperature of 270 K. If the temperature is decreased to 250 K, what will be the new volume?
3. A gas occupies a volume of 8 L at a temperature of 350 K. If the temperature increases to 500 K, what will be the new volume?
4. A gas has a volume of 5 L at 273 K. If the temperature is raised to 373 K, what will the volume be?
5. A gas occupies 3 L at a temperature of 280 K. If the temperature is increased to 350 K, what will be the new volume?

TEMPERATURE CONVERSION

**1. Convert the following units :**

- a. 100°C to K
- b. 150°C to K
- c. 780 K to $^{\circ}\text{C}$ to K
- d. 170 K to $^{\circ}\text{C}$